

Notes: TU IOE PYQ 2069–2082 (23 papers). Regular = **bold**, Back / New Back = normal. Single common paper for BEX + BCT — no programme split (**82 Bh** prints BCT/BEI). Marks in (); unique values only. Keys: (I)=Importance/Need (+)=Advantage (-)=Disadvantage +Fig=Figure +Eg=Example Vs=Compare/Differentiate.

1. Introduction

- Data mining +process / general steps; applications (2+3)(2+6)(1+5)(4)(10)
- KDD steps +Eg; KDD challenges that motivated data mining (2+5)(3+7)(3+3)
- “Data rich but information poor” / “data into knowledge” — justify (8)(6)
- Data warehousing +Eg where used; key features (2+4)(2+5)
- DW architecture +analytical processing (8)
- DW Vs data mining; DW Vs database / RDBMS (similar too) (3+7)(2+2)
- DM system integrated with DB/DW +Fig process (4+2)
- Data warehouse Vs data mart; snowflake scheme +Eg (2+4)(15)(5)

2. Data Preprocessing

- Data types of attributes; nominal Vs ordinal; symmetric Vs asymmetric data (10)(5)(6)
- Data pre-processing (I) +major tasks; data clearing; data sampling (2+7)(4+4)(5+5)(2+4+4)
- Data quality elements; data transformation by normalization +Eg (2+6)
- Normalize 200/300/400/600/1000 — min-max (0,1), Z-score, decimal scaling (2+2+2)
- Stop-word frequency (27 values, 13...70) — bin means depth 3, min-max, Z-score ($\sigma=12.94$), decimal scaling (6)
- Missing values — methods to handle (5)(2)(4+4)
- Data integration issues (5)
- Curse of dimensionality +avoid; noisy data impact (5)(6)
- Dimensionality reduction (I) (5)(4)
- PCA via dimensionality reduction; principal components + % variance (3×3 covariance) (10)(4)
- Short notes: data smoothing, data normalization, data transformation (3)(4)

OLAP & Multidimensional

- OLAP operations over multidim. DW; OLAP Vs OLTP (6+4)(10)(5+5)(2+5+3)
- Dimensional data; base & apex cuboid; slice & dice; roll up/down +Eg (2+3+3+3)
- Star schema (spectator/location/game, charge) + OLAP ops (3+3)
- Snowflake schema (time/location/supplier/brand/product) + OLAP ops (3+3)
- Short notes: OLAP; OLAP operations; OLAP and OLTP; OLAP cubes (4)(5)

Similarity Measures

- Similarity/dissimilarity; cosine + Euclidean (Size/Weight/Color/Taste table) (3+3+3)
- Types of similarity measures +application areas; why important (3)(2+6)
- Jaccard, SMC, symmetric dissimilarity (Ram/Laxmi/Shyam); cosine of d1,d2 (2+2+2+1)
- Manhattan & Supremum distance matrix; cosine D1/D2; Jaccard+SMC P/Q (2+1+2)
- Minkowski distance — metric properties; metric in instance based classifier (10)(4)(3)

3. Classification

- Classification stages +Fig block; precision Vs recall inverse (2+3+5)
- Classification as supervised learning; impurity measures (10)

Decision Tree

- ID3 tree (Age/Income/Student/Credit_Rating → Buy's Computer) (10)
- ID-3 tree (SN/Credit history/Debt/Collateral/Income → Credit Risk, 14 rows) +evaluate model (2+9+3)
- Information gain tree (Gender/Car/Travel cost/Income → Transport) +predict; ID3 root node only (8)(2+8)
- Mushroom (Shape/Color/Odor) — ID3 root +full tree (2+5)
- Decision tree + information gain for attribute selection +Eg (8)

- Gini index +Eg; overfitting in DT; homogeneousness measure (I) (6+4)(10)(8)
- ID3 algorithm (asked with TPR/FPR numerical) (2)
- Best root attribute; classifier accuracy (4+6)
- Hunt's algorithm + test conditions per attribute type (10)
- DT with Naive Bayes concept +Eg (10)

Rule Based

- Rule based classifier working +Eg; classification principles (7)(2+8)
- CN2 algorithm; “Accuracy” & “Laplace” rule-evaluation measures (9)
- Conflict resolution strategies (I) (2+4)
- Decision tree classifier Vs rule based classifier (8)

Nearest Neighbor

- KNN K=3 Euclidean (iris Sepal/Petal table) (3+6)
- NN classifier + issues + better classifier; when used +algorithm (1+3+4)(2+4)

Bayesian

- Naive Bayesian classification +Eg; algorithmic steps +strengths (8)(6)(3+7)
- Baye's theorem +use in classification; how Naive Bayes cuts complexity (12)
- Prior Vs posterior probability (3)
- Classify X=(Home Owner, Marital Status, Income \$120K) — defaulted borrower (6)
- Classify X=(youth, medium, student yes, fair) — buys computer (RID table) (10)
- Naive Bayes limitation; Bayesian Belief Network +Fig (heart disease CPTs) (3+5)(10)
- Laplacian correction (4)

ANN Classifier

- How ANN / neural net classifier works +algorithm +general considerations (8)(2+6)(5)(5)
- Feed-forward net +Fig — activation at h_1 ; $\partial E/\partial o$; $\partial E/\partial w_{12}$ (2+3+4)

Issues: Overfitting / Validation

- Confusion matrix → accuracy, error rate, sensitivity, specificity, precision, recall (10)(4)
- When accuracy fails +Eg; classification error, false alarm rate (3+5)
- ROC +TPR, FPR, precision (1+3+6)
- ID3 +TPR/FPR/Accuracy from confusion matrix (100/40/60/300) (2+6)
- Confusion matrix from predicted sequences (classifiers X, Y) (10)
- Holdout, random sampling, k-cross validation, bootstrap — mechanism +(+)/(-) (8)
- Overfitting problem; overfitting and ROC (4)

4. Association Analysis

- Association analysis (I) +market-basket; where applicable; association rules (I) +Eg (2+5)(2+6)(10)(6)
- Support & confidence (I) (10)(2)
- Pattern evaluation (I) — statistical interestingness measures +Eg (8)
- Market basket analysis (4)

Apriori

- Apriori principle +Eg; optimizing brute force; generate rules from frequent itemset (2+6)(10)(15)(7)
- Why 100% confidence uninteresting but 99% interesting (4)
- Apriori numerals (2+7)(4+5)(8)(12)(10)(2+8)(2+6)(5+3)(3+7)
 - min sup count 4, conf 80% (T100–T106, F/A/C/D/G...)
 - min sup 2 (10=ACD, 20=BCE, 30=ABCE, 40=BE)
 - sup 50%, conf 80% (6 txns A,B/A,D/A,C/B,E/B,D,E/A,E,C)
 - sup 2/9, conf 7/9 (fast food Order:1–9, M1–M5) +rules $X \wedge Y \rightarrow Z$
 - conf 60%, sup 30%, L=3 +F-List (T1–T6, A,B,C,T,M,P,D,K)
 - sup 20%, conf 80% (TID 1–9, M1–M5)
 - sup 50% (1=ACD, 2=BD, 3=ABCE, 4=BDF)
 - sup 60%, conf 80% (T1–T5, A,B,C,D,E,F...K)
 - conf 65%, sup 50% (1–4, MILK/BREAD/CAKE/BUTTER/EGG)
 - min_sup 20%, binary matrix T2–T8 $\times$ A1–A9 +rules at any confidence
 - sup & conf 50% (1=A,B,C / 2=A,C / 3=A,D / 4=B,E,F)

FP-Growth / FP-Tree

- FP-Growth algorithm +Eg; (+) over Apriori; frequent itemset; no candidate generation; short note (2+5)(8)(10)(2+6)(1+8)(6)(12)(5)
- FP-Tree construction numericals (4+4)(2+7)(2+8)
 - TID 01–05 (f,a,c,d,g,i,m,p ...)
 - min sup 60%, conf 80% (T100–T500, {M,O,N,K,E,Y}...)
 - sup 33.34%, conf 60% (HotDogs/Buns/Ketchup/Coke/Chips)
 - Camera/Laptop/Pen drive/Mobile/Earphone (T1–T6)
 - sup 30%, conf 75% (1–8, P,Q,R,S,T)

Categorical / Sequential / Subgraph

- Categorical attributes — handling +issues +Eg (3+3)(2+3+3)(4)
- Subsequences of $\langle \{2,3,5\}\{6,7,8\}\{9,1\}\{7,4\} \rangle$; 4-subsequences of $\langle \{1,3\}\{2\}\{2,3\}\{4\} \rangle$ (3+3)
- Subgraph pattern; candidate sub-graphs by edge-growing +Fig; sequential + sub-graph pattern +Eg (2)(3+3)(4+4)
- AprioriALL sequential pattern mining; sequential pattern (3)

5. Cluster Analysis

- Cluster analysis + applications + types of clusters; Vs classification (8)(5)(2+6)(3+4)
- Clustering as unsupervised learning; Bisecting K-means (10)
- Types of clustering methods; partition based methods (10)
- Contiguous cluster + algorithm (10)
- Internet-user segmentation — choose algorithm +validate cluster (2+6)

K-means

- K-means algorithm +limitation/demerits; optimal K strategy (2+6)(4+4)(4+6)(10)
- Hierarchical Vs partitioning clustering; clustering Vs classification (2+8)(4)
- K-means numericals (2+8)(5+3)(10)(8)(4+8)
 - K=2, 6 instances (1,2)(2.5,1)(3.5,1.5)(4,1)(3.5,2.5)(5,3)
 - K=2, A/B (1,2)(2.5,4.5)(4,6)(3.5,4)(4.5,5)(3,6)
 - A/B (1,1)(1.5,2)(3,4)(5,7)(3.5,5)(4.5,5)(3.5,4.5)
 - K=3, matrix X (10 pts), centers (6.2,3.2)(6.6,3.7)(6.5,3.0)
 - K=2, 6 instances (1,2)(1.5,1)(3.5,1.5)(4,3)(3.5,2.5)(6,4)
 - K=2 from centroid (2,3) — (2,3)(3,3)(6,8)(8,8)(7,5) +K-means problems
 - K=3 **Manhattan**, same 10 pts as columns, 3 iterations
 - 1-D {5,12,18,24,30,42,48}, centroids {5,12,18} +SSE

Hierarchical

- Hierarchical clustering +dendrogram; agglomerative Vs divisive +Eg

(2+7)(10)(6)

- Dendrogram numericals (8)(4+4)
 - p1–p6 distance matrix (0.24/0.22/0.37...)
 - complete-linkage, Euclidean, p1–p6 (x,y) coordinates
 - A–F distance matrix — single-link & complete-link

DBSCAN

- DBSCAN algorithm +Eg; core/border/noise points; handling noise; avoid issues (3+5)(8)(8+2)(7)(10)
- DBSCAN numerical — (2,10)(2,5)(8,4)(5,8)(7,5)(6,4)(1,2)(4,9), $e=2$, minPts=2 (3+5)
- Neighborhood quantification; Eps & MinPts empirically; Eps1<Eps2 subset proof (2+2+2)
- Density based cluster + algorithm; outliers (8)
- Density reachable / density connected; DBSCAN clustering (4)

Issues

- Cluster validity measures; cluster evaluation / validation; issues in clustering (10)(4)(3)(6)
- Measures to compare two clusters (6)

6. Anomaly / Fraud Detection

- Anomaly detection (I) +distance based (2+3)(8)(2+6)(4+3)(10)(5)(2+5)
- Anomaly schemes / approaches; compare 3 methods (2+2+4)(6)
- Statistical Vs proximity Vs density Vs cluster based — strengths & weaknesses (6)
- Likelihood approach for anomaly detection (8)
- Where applicable; importance in security (3+3)(5)
- Types of outlier +Eg; density based outlier detection (5+3)
- Issues in anomaly detection (2)(4)(2+3)
- False alarm rate & detection rate (P=0.01 / 0.99, 99% normal) (3)
- Nearest-Neighbor for anomaly detection (10)
- Clustering for anomaly detection; base rate fallacy +Eg; convex hull method (3)(15)(7)

7. Advanced Applications

- Web mining — structure/taxonomy/categories; challenges; www mining (3+5)(6)(8)(5)(4)
- Text mining key steps; page rank +algorithm (4+4)(5)(4)
- PageRank numerical — damping 0.85, 5 iterations, initial 0.25 +Fig (8)
- Multimedia mining; visual data mining / data visualization (6)(3)(4)
- Time series data mining +Eg (6)(5)(4)
- Seasonality in time series data (5)